VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Programme Accredited by NBA

Course: Programming in Python **L-T-P-C:** 2-0-2-3

Course Code: CSEL102

Branches: CSE, CSM, CAI, CAD, CIC, ECE, and EEE

QUESTION BANK

Module-1: Python Programming Fundamentals		
S.No	**Short Answer Questions**	**Marks**
1	List out the various styles of comments supported by the python, briefly explain about them in one line	2M
2	Explain the difference between 'is' and '==' with an example	2M
3	Describe the purpose of range() function.	2M

4	Evaluate the output of the following code
	a = [1, 3, 5, 7, 9, 11]
	val = 7
	x = 1
	for i in a:
	  if i == val:
	  print("Found at", x, "iteration")
	  break
	  x += 1
	  else:
	  print("not found").
		2M

5	Define indentation in Python? Give an example.	2M
6	Display the odd numbers from 1 to n where n is the value provided by the user	2M
7	Briefly explain about the dynamic typing in Python with an example.	2M
8	Write a Python statement to declare multiple variables in a single line and assign values to them. Give an example.	2M
9	Find error and correct it:	2M

	msg = "She said "Good Morning" to me" print(msg)	
10	Differentiate between input() and print() functions in Python.	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python program to determine the disaster relief fund provided to a family based on the type of house damage and annual income. The program should accept the type of damage as a character (F for fully damaged and P for partially damaged) and the annual income of the family. If the house is fully damaged and the annual income is less than ₹3,00,000, the relief amount is ₹2,00,000; otherwise, the relief amount is ₹1,00,000. If the house is partially damaged and the annual income is less than ₹3,00,000, the relief amount is ₹1,00,000; otherwise, the relief amount is ₹50,000. The program should calculate and display the relief amount accordingly.	8M
2	Develop a Python program to calculate the fine for overdue library books. The program should prompt the user to enter the type of book as a character, where r represents a reference book, g represents a general book, and c represents a children's book, along with the number of days the book has been delayed. Based on the category selected, the fine must be calculated according to the given rate structure, which is Rs. 20 per day for reference books, Rs. 15 per day for general books, and Rs. 10 per day for children's books. Finally, the program should compute and display the total fine amount payable.	8M
3	Implement a Python program to process loan applications for multiple applicants using a for loop. The program should first accept the number of applicants. For each applicant, input the applicant's name, age, annual income, and credit score. An applicant is eligible for a loan only if the age is between 21 and 60 (inclusive), the annual income is at least ₹3,00,000, and the credit score is 650 or above. If the applicant satisfies all the conditions, calculate the loan amount as three	8M

	times the annual income. Otherwise, reject the application. After processing all applicants, display the total number of eligible applicants and the total number of rejected applicants.	
4	Implement a Python program to determine whether a given integer is a prime number using an efficient approach. The program should accept an integer input from the user and first handle the special case where the number is less than or equal to 1 by displaying that it is not a prime number. For values greater than 1, the program should check for divisibility only from 2 up to the square root of the given number instead of checking all numbers up to n-1. If any divisor is found within this range, the program must display that the number is not prime; otherwise, it should display that the number is prime.	8M
5	Implement a Python program to accept a string from the user and perform the following operations: 1. Count the number of vowels and consonants 2. Display the string in reverse order 3. Check whether the given string is a palindrome or not	8M
6	Develop a Python program to accept a number n from the user and perform the following operations: 1. Display all numbers from 1 to n 2. Print only the numbers divisible by both 3 and 5 3. Count how many such numbers exist 4. Compute and display the sum of these numbers	8M

Module-2: Sequences, Map and Sets		
S.No	**Short Answer Questions**	**Marks**
1	Define list and how it is created	2M
2	Explain remove() and discard() in python set.	2M
3	List any four dictionary methods	2M
4	List out the difference between lists and tuples?	2M
5	Describe the purpose of slicing and demonstrate with an example.	2M

6	Evaluate the output of the following python script? t = (5) print(type(t))	2M
7	What is list comprehension in Python? Give a simple example.	2M
8	Differentiate between append() and extend() methods in Python lists with examples.	2M
9	How are keys and values accessed in a dictionary? Give an example.	2M
10	What is the difference between remove() and pop() methods in Python lists?	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python program that asks the user to input two strings of equal length. If the lengths are unequal, display a suitable message and terminate. If the lengths match, construct a new string by combining the lowercase version of characters from the first string and the uppercase version of characters from the second string at corresponding positions, and print the resulting string.	8M
2	Design a Python program that generates a list of 20 random numbers between 1 and 100. The program should print the list, calculate and display the average value, identify the highest and lowest numbers, determine whether any duplicate values exist in the list, count the number of elements greater than 50, and display the total count.	8M
3	A social media platform wants to analyze user comments and find out how many times each word appears in a comment. Users may type words in different formats such as “Like”, “like”, or “LIKE”, but the platform should treat all of them as the same word. Develop a Python program that reads a comment (sentence) from the user and displays the frequency of each word using a dictionary. The program should treat uppercase and lowercase words as the same word. After writing the program, explain how the dictionary helps in counting the words and what would happen if the	8M

	sentence is not converted into lowercase before counting.	
4	A movie streaming platform needs to record the daily viewing details of its users. Each record should contain the user's name and the number of hours watched in a day. Since the record should not be changed once created, it must be stored as a tuple. All user records should be stored inside a list. Implement a Python program to accept the viewing details of multiple users and store each record in the form (user_name, hours_watched). The program should display all user records, identify the user who watched the maximum number of hours, and calculate the total hours watched by all users.	8M
5	Write a Python program to manage student marks using a dictionary. The program should accept the names of students and their corresponding marks for a given number of students. The data must be stored in a dictionary where the student name acts as the key and the marks act as the value. After storing the data, the program should display all student records, identify and display the student who has secured the highest marks, and finally compute and display the average marks of all students.	8M
6	Develop a Python program using sets to perform various operations. The program should accept two sets of elements from the user and display the union, intersection, and difference of these sets. It should also determine whether one set is a subset of the other. Additionally, the program should accept a list containing duplicate elements and demonstrate how sets can be used to remove duplicates, displaying the resulting list of unique elements.	8M

Module-3: Functions		
S.No	**Short Answer Questions**	**Marks**
1	Define a function in Python. Write the syntax for defining a function.	2M
2	Differentiate between a function definition and a	2M

	function call with an example.	
3	What are default parameters in Python functions? Give an example.	2M
4	Explain keyword arguments in Python with an example.	2M
5	What are arbitrary arguments in Python functions? Distinguish between *args and **kwargs.	2M
6	What is variable scope in Python? List the different types of scope.	2M
7	What is the difference between local variables and global variables?	2M
8	Define higher-order functions in Python. Give one example.	2M
9	What is the purpose of the map() function? Provide a simple example.	2M
10	Differentiate between filter() and reduce() functions in Python.	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python program to process employee salary details using functions. The program should accept the name, basic salary, and number of years of experience of an employee. A function should be defined to calculate the bonus based on experience, where employees with more than 5 years of experience receive a 20% bonus and others receive a 10% bonus. Another function should compute the total salary including the bonus. The program should display the employee details along with the calculated total salary.	8M
2	Implement a Python program to demonstrate the use of functions with multiple parameters, default parameters, and keyword arguments. Define a function to calculate the total cost of purchasing items, where the function accepts the item price, quantity, and an optional discount percentage (default value 0). The function should compute and return the final amount after applying the discount. Demonstrate the function by calling it using both positional and keyword arguments.	8M
3	Design a Python program using arbitrary argument lists to process student marks. Define a	8M

	function that accepts a variable number of marks using *args, calculates the total and average, and returns the result. Extend the program to accept student details using **kwargs and display the complete information along with computed results.	
4	Develop a Python program to demonstrate scope rules. The program should define both global and local variables and show how variables behave inside and outside functions. Include examples to illustrate the use of the global keyword and explain how variable values change when modified within a function.	8M
5	Implement a Python program to demonstrate the use of higher-order functions. The program should accept a list of integers from the user and perform the following operations: use the map() function to square each number, use the filter() function to extract even numbers, and use the reduce() function to compute the sum of all elements. Display the results of each operation.	8M
6	Develop a Python program that uses functions to process a list of numbers. The program should define separate functions to find the maximum, minimum, and sum of elements in the list. The main program should call these functions and display the results. Additionally, modify the program to use a higher-order function to apply a transformation (such as doubling each element) to the list.	8M

Module-4: Object Oriented Programming		
S.No	**Short Answer Questions**	**Marks**
1	Define Object-Oriented Programming (OOP). List any two features of OOP.	2M
2	Define a class in Python? Give an example.	2M
3	Define object? How is an object created in Python?	2M
4	Differentiate between a method and a function in Python.	2M
5	Explain attributes in a class? Distinguish between instance and class attributes.	2M

6	Explain the different types of access specifiers in Python.	2M
7	What is inheritance? Give an example of a base class and a subclass.	2M
8	Define polymorphism in Python with a simple example.	2M
9	What is method overriding in Python?	2M
10	Describe the purpose of the self-keyword in Python classes?	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python program to manage student details using classes and objects. Define a class Student with attributes such as name, roll number, and marks. Include methods to accept data and display student details. Extend the program to calculate and display the grade based on marks using appropriate conditions.	8M
2	Write a Python program to demonstrate the use of class and instance attributes. Define a class Employee with a class attribute representing the company name and instance attributes such as employee name and salary. Include methods to display employee details and show how class attributes are shared among all objects.	8M
3	Design a Python program to demonstrate inheritance. Create a base class Person with attributes such as name and age. Derive a subclass Teacher that includes additional attributes such as subject and salary. Implement methods to display details of both classes and show how the subclass accesses the base class properties.	8M
4	Develop a Python program to illustrate method overriding in inheritance. Create a base class Shape with a method to calculate area. Derive subclasses such as Rectangle and Circle that override the area method based on their formulas. Display the area for different shapes.	8M
5	Write a Python program to demonstrate polymorphism using functions and objects. Define multiple classes such as Dog, Cat, and Bird, each having a method named sound(). Create objects of	8M

	these classes and demonstrate how the same method name behaves differently for different objects.	
6	Design a Python program to demonstrate single inheritance. Create a base class Person with attributes such as name and age. Derive a class Employee from Person that includes additional attributes such as employee ID and salary. Implement methods to accept and display details, and demonstrate how the subclass inherits and utilizes the properties of the base class.	8M

Module-5: File Handling		
S.No	**Short Answer Questions**	**Marks**
1	Define a file in Python? List different file modes used in file handling.	2M
2	Explain the difference between text files and binary files in Python	2M
3	Describe the purpose of open() function in Python? Write its syntax.	2M
4	Differentiate between read(), readline(), and readlines() methods?	2M
5	Explain Pandas Series? Give an example.	2M
6	Define a DataFrame in Pandas. How is it different from a Series?	2M
7	How can a CSV file be read using Pandas? Write the syntax.	2M
8	How do you write data from a DataFrame into a CSV file?	2M
9	Explain indexing in a Pandas Series? Give an example.	2M
10	Describe the purpose of head() and tail() methods in Pandas?	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python program to perform file handling operations. The program should accept text input from the user and write it into a file. After writing, the program should read the contents of the file and display them. Additionally, count and display the number of lines, words, and characters present in the file.	8M

2	Implement a Python program to demonstrate different file modes. The program should create a file, write data into it using write mode, append additional data using append mode, and finally read and display the complete contents of the file.	8M
3	Design a Python program using Pandas to create a Series from a list of numbers. The program should display the Series, perform basic operations such as finding the sum and mean, and demonstrate how to access elements using indexing.	8M
4	Develop a Python program to create a DataFrame from a dictionary containing student details such as name, age, and marks. The program should display the DataFrame, calculate the average marks, and identify the student with the highest marks.	8M
5	Design a Python program to read data from a CSV file using Pandas. The program should display the first five and last five records, show the column names, and compute basic statistics such as mean and maximum values for numerical columns.	8M
6	Implement a Python program to process data from a CSV file containing employee details. The program should read the file into a DataFrame, filter employees whose salary is above a given threshold, and display the filtered results. Additionally, the program should save the filtered data into a new CSV file.	8M
Module-6: Graphical User Interface		
S.No	**Short Answer Questions**	**Marks**
1	Define Tkinter? Mention its purpose in Python.	2M
2	List out the basic steps to create a simple Tkinter window.	2M
3	Explain about a widget in Tkinter? Give any two examples.	2M
4	Explain the purpose of the Label widget with an example.	2M
5	Differentiate between Entry and Text widgets?	2M
6	Define a Button widget? How is it used in Tkinter?	2M
7	Differentiate between Checkbutton and Radiobutton.	2M
8	Briefly describe Frame widget in Tkinter?	2M

9	List the different geometry managers in Tkinter and mention their purpose.	2M
10	Define event handling in Tkinter? Name any two types of events.	2M
S.No	**Essay Answer Questions**	**Marks**
1	Develop a Python GUI application using Tkinter to create a simple user registration form. The form should include widgets such as Label, Entry, and Button to accept user details like name and age. When the user clicks the submit button, the entered details should be displayed on the window.	8M
2	Implement a Python program using Tkinter to demonstrate the use of different widgets such as Label, Button, Entry, Checkbutton, and Radiobutton. The program should allow the user to select options and display the selected values when a button is clicked.	8M
3	Design a Tkinter-based application to create a basic calculator that performs addition, subtraction, multiplication, and division. The application should accept user input through Entry widgets and display the result when the corresponding operation button is clicked.	8M
4	Develop a Python program using Tkinter to demonstrate different geometry managers. The program should arrange multiple widgets using pack(), grid(), and place() and show how each manager positions the widgets differently.	8M
5	Implement a Tkinter program to create a Listbox that displays a list of items. The program should allow the user to select an item from the list and display the selected item when a button is clicked.	8M
6	Design a Python GUI application using Tkinter to demonstrate event handling. The program should respond to keyboard and mouse events such as key press, mouse click, and mouse movement, and display the corresponding event information on the window.	8M